

sold and serviced by a third-party. Each company involved in the chain specialises in its area of work and is able to provide the best possible product at an optimal cost. Fig. 2 shows the evolution of outsourcing (activities marked in blue are generally outsourced). This shows that every activity, except IP issues, can be outsourced. Companies may make an exception if their IPs are well protected (e.g., Apple Inc.).

The main reasons for this shift to outsourcing are:

**Time to market pressure.** To get products into the market in the shortest time possible in order to establish a lead and capture the market.

**Specialised expertise and capital-intensive nature of manufacturing.** With the usage of surface mount and other small footprint technologies, need for specialised technologies for manufacturing has become mandatory. This technology is capital intensive and needs process, experience and knowledge.

**Compliance to regulatory needs.** Increased environmental awareness is driving the product-development-and-manufacturing process to be environment friendly. This calls for specialised processes, which are forcing the shift to outsourcing because the equipment to comply to green laws are expensive and doing it in-house is not cost-effective.

**Changing ecosystem and the need to be agile and lean.** Extremely-short component life cycles and a highly-volatile market make it difficult to predict prices. A rugged supply chain and component engineering practice ensures that shortage or obsolescence of components does not impact production. This issue leads to fluctuating bill of material (BoM) prices, and many times they switch to untested, unreliable equivalents, which ultimately leads to failures in the field. Also, using components produced with surface mount technology (SMT) almost guarantees a reliable product.

**Markets going global.** Designing products for Indian markets only is not smart as a lot of those products have a ready market in the rest of the world. Also, country-specific laws are sometimes imposed as invisible trade barriers

## Different Activities of Each Phase of a Life Cycle and a Brief Description of Each Activity

| Phase         | Activity                         | Brief description                                                                 |
|---------------|----------------------------------|-----------------------------------------------------------------------------------|
| Concept       | PoC                              | Proof of concept (functional prototype)                                           |
|               | IP                               | Intellectual property (patents)                                                   |
| Design        | Improvement                      | Improving an existing product                                                     |
|               | Refresh                          | No physical changes but redesigning to save cost                                  |
|               | Redesign                         | New shape and other changes with reduced cost                                     |
|               | New design                       | Absolutely new product to market                                                  |
| Engineering   | DFA                              | Design for assembly                                                               |
|               | DFM                              | Design for manufacturing                                                          |
|               | OFT                              | Design for test                                                                   |
|               | DFC                              | Design for cost                                                                   |
|               | DFR                              | Design for reliability                                                            |
| Manufacturing | DFE                              | Design for environment                                                            |
|               | PCB assembly                     | Components soldered to PCB                                                        |
|               | Subsystem build                  | PCB assembly, and testing of PCB and additional elements but not the full product |
|               | Box build                        | Delivering a fully-tested product                                                 |
|               | Fulfillment                      | Delivering product to end user directly                                           |
| Support       | Manufacture support              | Providing support during manufacture of product                                   |
|               | Field failure analysis           | Failure analysis of components, parts and systems returned from field             |
|               | Test and repair                  | Testing and repair of post warranty product for OEM                               |
|               | Homologation                     | Changing products based on country-specific requirements                          |
|               | Cost reduction/value engineering | Continuous process to reduce cost of product                                      |
| Sustain       | Recycle or disposal              | Disassembly, recycling and disposal of product at the end of its life             |
|               | Bug Fix                          | Fixing software or hardware bugs                                                  |
|               | Enhancements                     | Enhancing the features with minimal changes                                       |
|               | Life extension                   | Supporting product throughout its life, even after discontinuation                |

(ITBs) to restrict imports. Similar products from competitors outside India meet global standards, which increase the competition in today's brand-conscious market.

**Customer expectations and experience.** The rising expectation level of users has redefined the need for a product as a service—any shortfall reflects badly on the product.

These are some of the trends that have changed the way business is carried out. For an SME to handle all these changes in-house can be challenging as these need expertise, special skills and investment in machinery. It also requires investment in human resources and

upgradation of the skills of employees. So, in order to meet challenges, outsourcing in full or in part is unavoidable.

Most SMEs today are cost centric and averse to the idea of outsourcing, which affects the image of the product and the OEM goes down as a whole. This invisible damage and the impact is felt by the OEM only when the end customer switches to another vendor's product. Today, OEMs have outsourced the support function to specialised partners.

## Challenges SMEs face

A changing environment brings with it a host of challenges. Some critical chal-